$$\boldsymbol{\Phi} = \boldsymbol{\chi}^{-1}\boldsymbol{\Psi}\,\boldsymbol{\chi}.$$

In such a case the iterates of the two maps will also be conjugate and, if they are numerical methods, they will have similar stability properties and performance (e.g. the same effective order). It is difficult to separate the relevance of the two properties in cases where symplectic and reversible maps are conjugate. This is however rarely the case and certainly does not hold generically for discrete maps in many dimensions [209]. There is no direct correspondence between reversible and symplectic maps, however each class of maps admits certain theorems of dynamical systems which are in many ways analogous (for example, the Kolmogorov-Arnold-Moser, or KAM, theory for symplectic maps near elliptic fixed points [386] has an analogue for reversible maps [97]).

Of particular importance for molecular dynamics are the following properties: a symplectic map will preserve volume, whereas a time-reversible map need not do so, and a symplectic integrator will approximately conserve energy due to the existence of the perturbed Hamiltonian, whereas a time-reversible integrator may give rise to a drift in energy [166].

3.7 Molecular Dynamics with Hard-Sphere Collisions

We close this chapter with a discussion of molecular dynamics with hard-core repulsion. This is a system which is not, properly speaking, Hamiltonian, but which has dynamics that are largely defined by a Hamiltonian system. The added complication is the nonsmooth character of the motion as impulses are applied at points of collision. The discussion below is a summary of work in [40, 184].

We assume, as usual a Hamiltonian $H = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})$ but we add the inequality constraint $\|\boldsymbol{q}_i - \boldsymbol{q}_j\| \geq \sigma_i + \sigma_j$ where σ_i, $i = 1, \ldots, \mathrm{N}$ is a core radius. The condition $\|\boldsymbol{q}_i - \boldsymbol{q}_j\| = \sigma_i + \sigma_j$, some i, j defines the constraint surface. When the particles are not touching, they move along Newtonian paths defined by the standard equations of motion. At impact, they exchange momentum and energy according to the rules of elastic collision. Specifically, at the point of contact, the momentum vectors of the two spheres are adjusted according to the rule:

$$\boldsymbol{p} := \boldsymbol{p} + \alpha\boldsymbol{u}_\perp$$

where $\boldsymbol{u}_\perp$ is normal to the constraint surface and α is a parameter chosen to maintain the conservation of energy. Let $\mathscr{R}_\mathrm{c}$ denote the action of the collision operator on the vector of positions and momenta. Then we can write the evolution formally as

$$\mathscr{F}_\tau^{\mathrm{h.s.}}(\boldsymbol{q},\boldsymbol{p}) = \mathscr{G}_{\Delta\tau_r} \circ \mathscr{R}_\mathrm{c} \circ \mathscr{G}_{\Delta\tau_{r-1}} \ldots \circ \mathscr{R}_\mathrm{c} \circ \mathscr{G}_{\Delta\tau_0}$$

where $\Delta\tau_1, \Delta\tau_2, \ldots, \Delta\tau_{r-1}$ are the times between collisions, $\Delta\tau_0$ is the time until the first collision, and $\Delta\tau_r$ is the time between the last collision and τ. Here $\mathscr{G}_t$ is the flow map of the smooth system (Hamiltonian H). The trajectory is thus piecewise smooth with continuous configurational path and momenta exhibiting finite jump discontinuities.

Various numerical methods have been proposed for collisional Hamiltonian systems [136, 176, 184, 263, 348, 352]. Typically, these schemes rely on the Verlet method to propagate the system between collisions, with collisions detected either (i) by checking for overlap at the end of the step, (ii) checking for overlap during the step, or (iii) approximating the time to collision before the step. Collisions lead to momentum exchange between particles according the principle outlined above.

3.7.1 Splitting Methods for Hard Sphere Molecular Dynamics

Splitting methods are suggested by considering a formal hard-sphere Hamiltonian

$$H_{\text{h.s.}} = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}/2 + U(\boldsymbol{q}) + U_{\text{h.s.}},$$

where $U_{\text{h.s.}}$ is assumed to be infinite for overlapping configurations (some $\|\boldsymbol{q}_i - \boldsymbol{q}_j\| \leq \sigma_i + \sigma_j$) and zero otherwise. The hard sphere system described by

$$H_{\text{free}} = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}/2 + U_{\text{h.s.}}$$

consists of purely ballistic (straight line) motion punctuated by momentum jumps. This system is formally integrable, i.e. we may write down its exact solution for any given initial conditions. (In practice, for large systems solved in parallel, this computation may be highly nontrivial [108, 336].) One approach is to consider the splitting $H = H_{\text{free}} + U$, evolving H_{free} for fixed intervals punctuated by impulses derived from the smooth potential U. In [40, 184] this algorithm is termed the "Primitive Splitting Algorithm" and can be described by the three steps:

$$\boldsymbol{P} := \boldsymbol{p} - \frac{h}{2} \nabla U(\boldsymbol{q}), \qquad (\boldsymbol{Q}, \boldsymbol{P}) := \mathscr{G}_h^{\text{free}}(\boldsymbol{q}, \boldsymbol{P}), \qquad \boldsymbol{P} := \boldsymbol{P} - \frac{h}{2} \nabla U(\boldsymbol{Q}).$$

Even with this symmetric form ("kick", "drift", "kick") where "drift" now involves the solution of the system H_{free}, it was shown in [184] that energy accumulates rapidly. Assuming a finite number of collisions on a fixed interval, the error behaves as $\mathcal{O}(h)$. In long simulations the energy error grows without bound.

To understand why this is the case, we consider a step of the primitive splitting algorithm applied to a simple system consisting of a fixed body at the origin and a single body (mass=1) moving under an external potential and in occasional contact with the fixed body, both bodies having radius 0.5. Let us consider three possibilities: (i) there is no collision in the timestep, (ii) there is a collision only at